PREFACE

THE present volume of the series "Library of Mathematical Handbooks"† is devoted to the so-called *method of statistical trials* (or *Monte Carlo Method*). In contrast to the volumes published previously, which were devoted to the classical branches of mathematics with a strictly delimited range of problems, an established terminology and a tradition of exposition, the mathematical methods considered in this volume have been developed only during the past 13 years.‡

These methods, which are applicable to the most widely diverse branches of computational mathematics, are linked by an idea which is common to them all.

Their basis lies in simulating statistical experiments by means of computational techniques, and recording numerical characteristics obtained from these experiments. Hence all of these methods have the general title of *method of statistical trials*, or *Monte Carlo method*. The solution of numerical problems by this method is closer in spirit to physical experimentation than to the classical numerical methods.

Indeed, the error of the Monte Carlo method cannot be estimated sufficiently well *a priori*, and as a rule it is found by means of investigating the standard deviation of the quantities being simulated. The solution might not be reproducible in detail. The solution is stable with respect to isolated errors in the operation of the computer which is employed.

The purpose of this book is to expound the basic features of the method, to give an adequate account of the techniques used in the Monte Carlo method and of typical examples, and to illustrate the principal fields of application of the Monte Carlo method. This objective explains the structure of the book.

In Chapter I (written by Yu. A. Shreider) the basic features of

† Publisher's Note: Tis is a reference to the series of Russian mathematics books.

‡ Essentially, we may speak of the application of the Monte Carlo method only after the publication of the work of N. Metropolis and S. Ulam in 1949 [203]. The term "Monte Carlo" first appeared in that work.

the method are explained and typical examples are considered of its application to simple problems in computational mathematics.

Chapter II (written by I. M. Sobol') contains a very detailed study of the computation of multi-dimensional integrals, which are (in a certain sense) a fundamental field of application of the method.

A variety of examples of statistical modelling of integrals is analysed therein, and the accuracy of the computations is investigated.

Chapter III (written by I. M. Sobol') and Chapter V (written by V. G. Sragovich) are devoted to the applications of the Monte Carlo method in those branches of physics and technology where it has been widely and successfully used. In fact, the Monte Carlo method was devised for solving problems in neutron physics. This is because the random processes which are investigated by the method (the transmission of beams of neutrons through scattering media, fluctuation of radio signals, etc.) are so complicated that their analytical solution is practically impossible. But the statistical simulation of such processes on a digital computer enables successful investigations to be made. These processes have specific peculiarities and many special devices were employed for modelling them, so that it proved to be convenient to divide them into two separate chapters.

Chapter IV (written by N. P. Buslenko) is devoted to the application of the Monte Carlo method to the investigation of servicing processes. This branch, which has been developed during the last few years, is concerned with the simulation of complex systems of control and operations research, and has great prospects in mathematical cybernetics.

In Chapters VI (written by D. I. Golenko) and VII (§§ 1, 2, 3 and 5 written by N. P. Buslenko, § 6 written by V. G. Sragovich, and § 4 written jointly by N. P. Buslenko and V. G. Sragovich) methods are considered for organizing statistical experiments on universal digital computers. The questions considered in these chapters are essential for any application of the method of statistical trials.

Two illuminating examples yet remain of fields of application of the Monte Carlo method: the investigation of learning machines and the study of the reliability of complicated electronic apparatus (e.g. computers). However, these and some other applications are still in the early stages of development, so that it would be difficult to include the corresponding material in a handbook. Questions relating to the first of them were partially clarified in the book

Stochastic Models of Learning, by R. Bush and F. Mosteller, a Russian translation of which was published by Fizmatgiz in 1962.

In view of the fact that the bibliographies for the various chapters overlap considerably, and also for convenience in handling, a general combined bibliography of the Monte Carlo method (assembled by I. M. Sobol') is given at the end of the book. All references in the text refer to this bibliography. Literature of a more general character, which is not concerned directly with the Monte Carlo method, is given after this bibliography, sharing a common numeration with it.†

General questions concerning the Monte Carlo method were discussed in a recently published book by N. P. Buslenko and Yu. A. Shreider [11]. The present volume differs from it giving a more detailed treatment of the material, and by the display of a wider range of applications.

We do not consider in this handbook any questions concerning the peculiarities of the structure of electronic digital computers which may be employed for the solution of problems by the Monte Carlo method. In particular, we do not consider questions concerning the construction of specialized computers.

A large group of authors collaborated in the writing of this handbook. Notwithstanding the extensive work which was done to harmonize the separate chapters, there remain a few inevitable discrepancies in the contents and in the style of presentation.

This handbook is designed for a wide circle of readers, ranging from those who are interested in the fundamental applications of the method, to those who are concerned with comparatively limited problems of the peculiarities of simulating physical processes. If we do not count Chapters III and V, which are intended for readers who are familiar with certain general facts concerning neutron physics and radio technology, then an understanding of the major part of the material expounded in this book requires from the reader a mathematical basis equivalent to a college course of mathematical analysis, and also a knowledge of the elements of the theory of probability. It is assumed that the reader is familiar with the basic facts concerning random events and quantities and their stochastic characteristics (the probability of the occurrence

† No such bibliography of more general literature appears in the Russian original. [G. J. T.]

of an event, mathematical expectation, variance). Moreover, it is necessary to have some knowledge of the law of normal distribution, Lyapunov's theorem and, for certain chapters, of the elements of mathematical statistics. It is also most desirable to be familiar with Markov processes.

The authors trust that this handbook will prove to be useful to those people who are interested in the Monte Carlo method and its applications; and, what is no less important, that it will introduce many readers to the idea of the usefulness of applying the Monte Carlo method for solving their problems. The authors express their gratitude to V. D. Rozenknop, for having given much valuable advice concerning the form of the text.

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